

The Kernel-Pair Root and Its Limits

A Hostile Universality Attack on Resolution Geometry, and the Typed Theorem Ladder
That Survives

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Abstract

A proposed compression claims that quotient, ignorance, saturation, assembly, Fisher/covariance structure, and refinement are jet-order shadows of one root, the kernel pair of an observation map, while cone and scale are declared on the quotient. We subject this claim to analytic and computational attack. The exact kernel pair does force observational equivalence, the set-level quotient, fibre saturation, and universal factorization of fibre-constant constructions. The stronger compression fails in six independent ways. At singular points $\ker dF_x$ need not be the tangent space of the exact fibre. Higher observational contact is controlled by jets of F , not by covariance, while smooth formal flatness need not imply exact local constancy. Fisher geometry requires a distinguishability structure and does not reconstruct covariance completions. Identical identifiability kernel pairs need not imply Blackwell equivalence. A sound declared groupoid need not generate all observation fibres. Precision and representation act on evaluation and downstream constructions rather than on the exact observation kernel. The surviving result is a typed root-and-extensions theorem: the kernel pair is universal only for exact extensional information; formal fibre jets, statistical comparison, metric detection, groupoid assembly, admissible cone, scale, transport, and fixed-protocol maximality are separate modules with explicit gates. A deterministic battery with master seed 20260912 passes all 15/15 registered attacks and emits machine-readable metadata. The computation is not a proof; it supplies finite witnesses, type checks, and regression artifacts for the theorem ladder.

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1 The attempted compression

Let $F : X \rightarrow Y$ be a declared observation map. Its kernel pair is

$$R_F = X \times_Y X = \{(x, x') \in X \times X : F(x) = F(x')\} \rightrightarrows X.$$

The attempted universal identification is attractive: R_F would be the exact ignorance relation; $\ker dF$ its first tangent shadow; a covariance or Fisher object its second-order shadow; X/R_F the observable quotient; and Blackwell refinement the order induced by coarsening kernel pairs. Assembly would compare R_F with a declared automorphism groupoid, while cone and scale would be declared on the quotient.

The compression contains a true universal core, but it crosses type boundaries. Exact equality, infinitesimal contact, statistical comparison, numerical evaluation, model representation, and physical geometric decoration are not one construction at different orders. The purpose of this paper is elimination: identify the maximum statement that remains valid after hostile counterexamples.

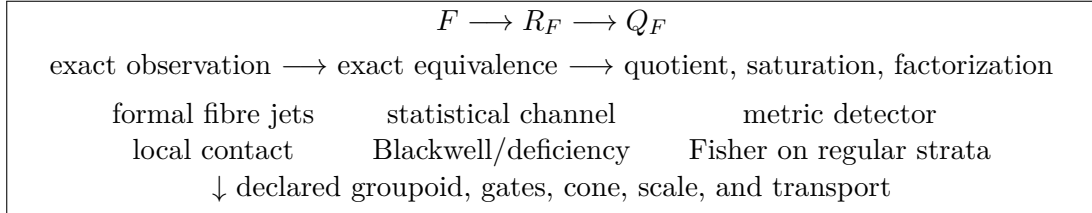


Figure 1: The surviving architecture. Only the first line is forced by the exact kernel pair.

2 Exact root: what the kernel pair really forces

Definition 2.1 (Exact observational equivalence). For $F : X \rightarrow Y$, define $x \sim_F x'$ iff $F(x) = F(x')$. The observable quotient is the set $Q_F = X/\sim_F$ with quotient map $q_F : X \rightarrow Q_F$.

Theorem 2.2 (Kernel-pair quotient universality). *Let Z be a set and $h : X \rightarrow Z$. The following are equivalent:*

1. h is constant on each F -fibre;
2. $R_F \subseteq R_h$;
3. there is a unique $\bar{h} : Q_F \rightarrow Z$ with $h = \bar{h} \circ q_F$.

Moreover, Q_F is canonically bijective to $\text{im } F$ as a set.

Proof. If h is fibre-constant, define $\bar{h}([x]) = h(x)$. Fibre constancy makes the definition independent of the representative. Surjectivity of q_F gives uniqueness. The other implications are immediate. The map $[x] \mapsto F(x)$ is a well-defined bijection from Q_F to $\text{im } F$. $\square$

Corollary 2.3 (Saturation). *A subset $A \subseteq X$ is observationally saturated iff it is a union of F -fibres, equivalently iff $A = q_F^{-1}(B)$ for a unique subset $B \subseteq Q_F$.*

Remark 2.4 (Category-sensitive quotient). The bijection $Q_F \cong \text{im } F$ is set-theoretic. Equality as topological, smooth, measured, stratified, or statistical objects requires additional hypotheses and the appropriate quotient structure. The exact universal claim must therefore name its category.

Proposition 2.5 (Kernel groupoid). *$R_F \rightrightarrows X$ is an equivalence-relation groupoid. Its orbits are exactly the observation fibres and its orbit set is Q_F .*

Proof. Source, target, identity, inverse, and composition are inherited from pairs (x, x') with equal F -value. The orbit of x is precisely $F^{-1}(F(x))$. $\square$

3 Attack I: differential kernels and singular fibres

Proposition 3.1 (Regular tangent identification). *If F has constant rank near x , then the fibre through x is an embedded submanifold near x and*

$$T_x F^{-1}(F(x)) = \ker dF_x.$$

Proof. This is the constant-rank theorem applied in adapted local coordinates. $\square$

Proposition 3.2 (Singular failure). *Without regularity, $\ker dF_x$ need not equal the tangent space of the exact fibre.*

Proof. Take $F : \mathbb{R} \rightarrow \mathbb{R}$, $F(t) = t^2$, at $t = 0$. Then $dF_0 = 0$, so $\ker dF_0 = \mathbb{R}$. But $F^{-1}(0) = \{0\}$, whose ordinary tangent space is $\{0\}$. $\square$

Thus $\ker dF_x$ is always a first-order infinitesimal null, but it is the tangent to the exact fibre only on suitable regular strata. Singular RG requires tangent cones, slices, or other stratified replacements, consistent with the orbit-cone architecture.

4 Attack II: higher jets are not Fisher covariance

Definition 4.1 (Order of observational contact). For a smooth curve γ with $\gamma(0) = x$, define

$$\nu_F(\gamma) = \inf\{r \geq 1 : (F \circ \gamma)^{(r)}(0) \neq 0\},$$

with $\nu_F(\gamma) = \infty$ when every derivative vanishes.

Definition 4.2 (Formal fibre-contact tower). Let

$$K_x^{[r]}(F) = \{j_x^r \gamma : j_0^r(F \circ \gamma) = j_0^r(\text{const}_{F(x)})\}.$$

Then $K_x^{[r+1]}(F)$ projects into $K_x^{[r]}(F)$, giving an order-by-order formal fibre tower.

For $v = \dot{\gamma}(0)$ and $a = \ddot{\gamma}(0)$, second-order contact requires

$$dF_x v = 0, \quad dF_x a + d^2 F_x(v, v) = 0.$$

The second condition includes acceleration data and the Hessian of F . It is not a covariance block and is not determined by a tangent vector alone.

Proposition 4.3 (Fisher is a first-derivative metric detector). *Let h be a positive-definite bilinear form on observable tangent directions and define $g^F = F^*h$. Then*

$$\text{rad } g_x^F = \ker dF_x.$$

For a Gaussian observation model with mean Jacobian J and fixed positive covariance Σ , this becomes $g^F = J^\top \Sigma^{-1} J$.

Proof. $g_x^F(v, v) = h(dF_x v, dF_x v)$ vanishes exactly when $dF_x v = 0$. The Gaussian formula is the standard pullback in coordinates. $\square$

The bilinear order of a metric must not be confused with jet order. At $F(t) = t^2$, ordinary first-order Fisher pullback at 0 vanishes while $d^2 F_0 = 2$ detects second-order variation.

Proposition 4.4 (Formal infinity is not exact infinity in the smooth category). *There exists a smooth map whose derivatives of every order vanish at a point although the map is not locally constant there.*

Proof. Set $F(0) = 0$ and $F(t) = e^{-1/t^2}$ for $t \neq 0$. Every derivative at 0 vanishes, but $F(t) > 0$ for every $t \neq 0$. Hence infinite formal contact does not equal an exact fibre germ without analyticity or a comparable effectivity hypothesis. $\square$

5 Attack III: Fisher and covariance are not kernel-pair data

The kernel pair records equality of observations. It contains no metric normalization, noise law, covariance completion, or divergence. Even after a Fisher metric is supplied, reconstruction remains many-to-one.

Proposition 5.1 (Covariance-completion nonuniqueness). *Let an observed three-dimensional Gaussian block have covariance $\Sigma_R \succ 0$, and append an unobserved independent block with covariance sI_3 . For mean parameters acting only in the observed block, Fisher information is Σ_R^{-1} for every $s > 0$, while the full covariance varies with s .*

Proof. The score and its covariance involve only the observed block. The null-block covariance is annihilated by the parameter-to-observation map and therefore does not enter the Fisher pullback. $\square$

This statement is weaker and better typed than calling full covariance a “second-order kernel-pair shadow.” Full covariance is extra statistical data; Fisher is a contraction of selected data on a regular model.

6 Attack IV: kernel-pair order is not the Blackwell order

Definition 6.1 (Blackwell comparison). For experiments $\mathcal{E} = (P_\theta)$ and $\mathcal{E}' = (Q_\theta)$ on the same parameter set, write $\mathcal{E} \succeq_B \mathcal{E}'$ if a parameter-independent Markov kernel K satisfies $Q_\theta = K P_\theta$ for every θ .

Proposition 6.2 (Deterministic monotone shadow). *If $H = G \circ F$, then*

$$R_F \subseteq R_H, \quad \ker dF_x \subseteq \ker dH_x$$

where differentials exist.

Proof. Equality of F -values implies equality after G . The differential inclusion follows from $dH_x = dG_{F(x)} \circ dF_x$. $\square$

Theorem 6.3 (Identifiability does not classify Blackwell information). *Two experiments may have the same parameter-side equality-of-law kernel pair but different Blackwell information.*

Proof. Consider $Y \mid \theta \sim N(\theta, 1)$ and $Z \mid \theta \sim N(\theta, 4)$. Both are identifiable, so each kernel pair is the diagonal $\theta = \theta'$. Yet Z is obtained from Y by adding independent $N(0, 3)$ noise. Their Fisher informations for the location parameter are 1 and $1/4$. Thus the kernel-pair map is a coarse monotone shadow, not a complete invariant of Blackwell comparison. $\square$

7 Attack V: precision and representation are different layers

Theorem 7.1 (Three-layer separation). *Fix an exact observation map F . Then:*

1. *a sensor or map null belongs to $\ker dF$ or, globally, to an exact F -fibre;*
2. *a precision or sampling null belongs to a thresholded numerical evaluator or estimator and may occur even when $dF(v) \neq 0$;*
3. *a representation null belongs to a downstream construction A through $\ker d(A \circ F)$ and may occur even when $dF(v) \neq 0$.*

Neither (b) nor (c) changes the exact kernel pair of the fixed F .

Proof. For (b), $\text{diag}(1, 10^{-12}, 0)$ has exact algebraic rank 2 but rank 1 under a 10^{-10} threshold. For (c), observations containing both x and x^2 retain quadratic dependence, while a downstream linear fit discards it. Both modify how available variation is treated, not equality of the exact observations. $\square$

The sensor family may itself refine. For a compatible directed system $F_S = q_{S'S} \circ F_{S'}$, define the observation limit F_∞ when the relevant inverse limit exists and

$$I_x = \ker dF_{\infty, x} = \bigcap_S \ker dF_{S, x}.$$

In finite-dimensional tangent spaces a descending family of kernels stabilizes in dimension under the directedness hypothesis. Incompatible measurement contexts, however, need not admit one jointly realizable product observation. A global limit is therefore relative to a compatible context.

8 Attack VI: assembly, cone, and scale are declared extensions

Let $\mathcal{A} \rightrightarrows X$ be a declared automorphism groupoid.

Definition 8.1 (Gauge soundness and completeness). $\mathcal{A}$ is sound when every arrow of $\mathcal{A}$ lies in R_F . It is orbit-complete when $\mathcal{A} \cdot x = F^{-1}(F(x))$ for all x .

Theorem 8.2 (Orbit-quotient comparison). *Gauge soundness induces a canonical surjection*

$$X/\mathcal{A} \longrightarrow Q_F, \quad [x]_{\mathcal{A}} \longmapsto [x]_F.$$

It is bijective exactly when $\mathcal{A}$ is orbit-complete.

Proof. Soundness gives representative independence. Surjectivity is immediate. Injectivity is equivalent to every pair in one F -fibre lying in one $\mathcal{A}$ -orbit. $\square$

Thus assembly is a comparison between a forced kernel groupoid and declared gauge data. It is not generated by the kernel pair alone.

Proposition 8.3 (Cone-scale separation). *On a regular quotient stratum of dimension at least three, a nondegenerate Lorentzian quadratic null cone determines a positive conformal ray $[g] = \{\Omega^2 g : \Omega > 0\}$, not a metric representative.*

Proof. Nondegenerate Lorentzian quadratic forms with the same null cone are pointwise proportional; time orientation and sign select the positive ray. Multiplying by a positive function leaves the null cone unchanged. $\square$

A cone is not forced to be quadratic, and an arbitrary reachability preorder need not determine Lorentzian geometry. Quadraticity, projectability across full observation fibres, and integrable scale calibration are separate gates.

9 Fixed-protocol universality after the attack

The universality claim survives only after the input is bundled with the protocol that supplies all non-kernel-pair choices.

Definition 9.1 (Declared protocol). Let

$$\Gamma = (h, G, \tau, N, S, C, T, \mathcal{A}, \mathcal{C}, \sigma)$$

collect, as applicable, a distinguishability module h , comparison metric G , threshold τ , nuisance convention N , stability gate S , conditioning/coarse-graining gate C , transport/gluing data T , declared groupoid $\mathcal{A}$, cone data $\mathcal{C}$, and scale calibration σ .

Theorem 9.2 (Typed kernel-pair root theorem). *For a declared observation system (F, Γ) :*

1. R_F uniquely determines exact observational equivalence, the set quotient Q_F , saturation, and universal fibre-constant factorization.
2. On regular strata, $\ker dF$ is the tangent fibre shadow. At singular points it is only the first formal null.
3. Higher local contact is represented by formal fibre jets of F , not by covariance or Fisher information.
4. A nondegenerate declared output distinguishability form pulls back to a metric whose radical is $\ker dF$ on its regular domain.
5. Deterministic garbling coarsens R_F , but stochastic Blackwell comparison contains strictly more information than the identifiability kernel pair.
6. A declared groupoid reproduces Q_F exactly only under gauge soundness and orbit completeness.
7. Precision and sampling act on evaluation; representation acts through downstream maps; neither is another exact kernel-pair order.
8. Cone, scale, anchor, transport, and assembly are induced or declared structures on or over Q_F , subject to their own descent and regularity gates.

Proof. Items (1)–(8) are respectively Theorem 2.2, Proposition 3.1 with Proposition 3.2, the fibre-contact construction, metric pullback nondegeneracy, Theorem 6.3, orbit-quotient comparison, Theorem 7.1, and the separate descent/reconstruction hypotheses above. $\square$

Theorem 9.3 (Fixed-protocol theorem ladder). *Assume a regular finite classical protocol fibre in which:*

1. *equality of laws defines the extensional quotient;*
2. *a normalized Fisher–Rao module is fixed under its stated invariance hypotheses;*
3. *$G \succ 0$, $L = G^{-1}F_I$ is G -self-adjoint, and $\tau \notin \text{spec } L$;*
4. *the declared gates admit a greatest resolved subobject;*
5. *transports satisfy identity, naturality, and cocycle laws.*

Then the ladder

$$(E, \Gamma) \rightarrow Q_E \rightarrow g_F \rightarrow P_\tau = \mathbf{1}_{(\tau, \infty)}(L) \rightarrow W_{\text{adm}} = \max \text{Adm}_\Gamma(\text{im } P_\tau) \rightarrow \{T_{ij}\}$$

is unique up to protocol-preserving natural isomorphism and declared metric normalization. No canonical protocol-free section $E \mapsto \Gamma$ follows.

Proof. Quotient universality fixes Q_E . The imported metric characterization fixes the regular classical metric up to declared normalization. Spectral functional calculus fixes P_τ . Soundness plus completeness fixes the greatest admissible subobject. Naturality and cocycle closure fix compatible transported copies. Every remaining choice appears explicitly in Γ . $\square$

Corollary 9.4 (Exact locus of universality). *The quotient is universal across extensional constructions of a fixed experiment. The final admissible sector is greatest only within a fixed protocol fibre. Therefore “RG is universal” is defensible only as a modular, typed, conditional statement.*

10 Hostile computational battery

10.1 Design

The accompanying Python artifact performs fifteen deterministic checks. Each check targets a proposed collapse or a surviving gate. The battery uses a master seed and named subseeds to prevent accidental coupling between randomized blocks. Analytic witnesses are retained even where randomness is unnecessary.

Attack	Recorded result	Forced conclusion
Regular linear fibre	error 4.97×10^{-16} , nullity 3	Regular linear kernel moves stay in one fibre.
Singular fibre	$\dim \ker dF = 1$, fibre tangent dimension 0	Tangent equality requires regularity.
Second jet	$dF(0) = 0$, $d^2F(0) = 2$	Higher contact belongs to jets of F .
Smooth flatness	positive off-origin samples	Formal infinite order need not be exact fibre constancy.
Covariance completion	Fisher difference 0; covariance traces 13.0518, 37.0518	Fisher does not reconstruct the full covariance completion.

Attack	Recorded result	Forced conclusion
Gaussian garbling	empirical variances 0.9980, 3.9877; Fisher 1, 1/4	Same diagonal kernel pair does not give Blackwell equivalence.
Deterministic garbling	nullity $3 \rightarrow 4$	Post-processing enlarges the differential null.
Precision layer	exact rank 2, thresholded rank 1	Numerical null is evaluator-relative.
Representation layer	residual $9.4517 \rightarrow 1.17 \times 10^{-14}$	A richer downstream class recovers retained data.
Gauge completeness	10 declared orbits, 5 fibres, 5 completed orbits	Soundness alone does not imply orbit equality.
Measurement compatibility	Pauli commutator norm 2.8284	A single global joint refinement requires compatibility.
Cone versus scale	common null residual 3.55×10^{-15} at scale 7.3	Cone fixes a conformal ray, not scale.
Protocol-free uniqueness	two thresholds give different projectors	The experiment does not select the protocol.
Fixed-protocol maximality	4 sound sectors, 1 complete sector	Completeness restores uniqueness inside the declared fibre.
Fisher type check	$(3 \times 6)^\top (3 \times 3)(3 \times 6) = 6 \times 6$	Fisher acts on parameter tangents.

Proposition 10.1 (Battery outcome). *With master seed 20260912, all 15/15 registered checks pass. The result file records the environment, each observed value, the consequence label, and a SHA-256 digest of the canonical result payload before insertion of the digest field.*

Remark 10.2 (Epistemic role). The battery is not a proof of Theorems 9.2 or 9.3. It supplies executable finite witnesses, catches dimensional mistakes such as an invalid post-processing matrix, and acts as a regression suite for future reformulations.

11 Reproducibility

11.1 Artifacts

The complete deterministic artifacts are:

- `kernel_pair_universality_attack.py`;
- `kernel_pair_universality_attack.json`;
- `kernel_pair_universality_attack.log`;

11.2 Seeds and numerical contract

Role	Seed
Master battery	20260912
Regular linear fibre	2026091201
Fisher completion	2026091202
Gaussian Blackwell witness	2026091203
Reserved oblique-projector stream	2026091204

The default absolute tolerance is 10^{-10} . The Gaussian check uses 200000 standard samples and acceptance windows declared directly in the script. Exhaustive finite masks and analytic examples are seed-independent. The seed controls only generated matrices and Monte Carlo sanity checks.

11.3 Execution

`python3 kernel_pair_universality_attack.py`

A successful run exits with code zero and reports **checks:** 15/15. Reproduction should compare structured JSON fields rather than require byte-identical floating-point text across all platforms.

12 Non-claims and falsification criteria

Non-claim 12.1 (No one-object collapse). The paper does not claim that quotient, Fisher metric, covariance, Blackwell comparison, assembly, cone, scale, and representation are all recoverable from one kernel pair.

Non-claim 12.2 (No unrestricted tangent equality). $\ker dF_x = T_x F^{-1}(F(x))$ is asserted only under suitable regularity. Singular points require a different tangent notion or stratified model.

Non-claim 12.3 (No smooth jet completion theorem). Infinite formal fibre contact does not imply exact local fibre constancy in the C^∞ category. Analytic or formal-effectivity hypotheses must be stated when such an implication is used.

Non-claim 12.4 (No Fisher reconstruction of the experiment). Fisher information does not determine the full likelihood family, covariance completion, higher likelihood derivatives, or Blackwell class.

Non-claim 12.5 (No protocol-free uniqueness). A bare experiment does not select G , τ , nuisance conventions, admissibility gates, groupoid, cone, scale, connection, or transport data.

Non-claim 12.6 (No global compatible observation by default). A total observation limit is asserted only for a compatible jointly refinable sensor family. Incompatible measurement contexts are treated contextwise.

Non-claim 12.7 (No automatic Lorentzian realization). A descended cone need not be quadratic, and a quadratic cone fixes only a conformal class. No spacetime ontology, field equation, stress tensor, or physical clock identity is derived.

Non-claim 12.8 (No proof by battery). Passing the finite attack validates the implementation and witnesses independence. It does not replace the analytic proofs.

The theorem ladder must be revised if any of the following occurs: an extensional construction varies inside one equality-of-law class; two nonisomorphic quotients satisfy the same stated universal property in the same category; a claimed regular fibre violates Proposition 3.1; a

metric radical differs from $\ker dF$ despite nondegeneracy on observable directions; a kernel-pair invariant is shown to classify Blackwell comparison; two distinct greatest sectors exist for one fixed admissibility family; or declared transport fails identity or cocycle closure.

13 Conclusion

The unification attempt succeeds at one exact root and fails as a total compression. The observational kernel pair is the canonical source of exact equivalence. It forces the observable quotient, fibre saturation, and universal factorization. Its regular first-order shadow is $\ker dF$; its genuine higher-order shadow is the formal fibre-jet tower. Fisher geometry is not that tower. It is a separately grounded metric detector using first derivatives and a declared statistical distinguishability structure. Blackwell informativeness is not the kernel-pair order. It refines that coarse identifiability shadow through channel structure. Assembly compares the forced kernel groupoid with a declared groupoid and requires completeness for equality. Precision and representation occupy evaluator and downstream layers. Cone, scale, anchor, transport, and physical realization remain explicit modules.

The final defensible statement is therefore:

$$\boxed{\text{kernel pair} \Rightarrow \{\text{exact quotient, ignorance, saturation, factorization}\}}$$

and, only with stated gates,

$$\boxed{\begin{array}{l} \text{formal fibre jets} \Rightarrow \text{local contact order,} \\ \text{statistical/channel structure} \Rightarrow \text{Fisher and Blackwell modules,} \\ \text{declared protocol} \Rightarrow \text{maximal sector, assembly, cone, scale, transport.} \end{array}}$$

This is weaker than universal derivation from observation alone, stronger than an arbitrary construction, and exactly aligned with fibrewise canonicity: after the complete experiment and protocol are declared, no undeclared choice remains inside the theorem's domain.

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